

My Solutions to Problem Sheet 2 QFT by David Tong

Baset

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1.

According to the definitions made:

$$\begin{aligned} [a_n, a_m] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m + \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{i[q_n, p_m] + i[p_n, q_m]\} = 0 \end{aligned} \quad (1)$$

$$\begin{aligned} [a_n, a_m] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m + \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{i[q_n, p_m] + i[p_n, q_m]\} = 0 \end{aligned} \quad (2)$$

$$\begin{aligned} [a_n^\dagger, a_m^\dagger] &= [\sqrt{\frac{\omega_n}{2}}q_n - \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m - \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{-i[q_n, p_m] - i[p_n, q_m]\} = 0 \end{aligned} \quad (3)$$

$$\begin{aligned} [a_n, a_m^\dagger] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m - \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{-i[q_n, p_m] + i[p_n, q_m]\} = \delta_{nm} \end{aligned} \quad (4)$$

Using 4:

$$\begin{aligned} H &= \sum_n \frac{1}{2}\omega_n(a_n a_n^\dagger + a_n^\dagger a_n) = \sum_n \omega_n(a_n^\dagger a_n + \frac{1}{2}) \\ &= \sum_n \omega_n((\sqrt{\frac{\omega_n}{2}}q_n - \frac{i}{\sqrt{2\omega_n}}p_n)(\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n) + \frac{1}{2}) \\ &= \sum_n \omega_n(\frac{1}{2}\omega_n q_n^2 + \frac{1}{2\omega_n}p_n^2 + \frac{i}{2}[q_n, p_n] + \frac{1}{2}) \\ &= \sum_n \frac{1}{2}(p_n^2 + \omega_n^2 q_n^2) \end{aligned} \quad (5)$$

If we use the first line of the above calculation as H :

$$H|0\rangle = \sum_n \omega_n (a_n^\dagger a_n + \frac{1}{2}) |0\rangle = \infty |0\rangle \quad (6)$$

In the above H terms, the only term that produces a non-zero energy when acting on vacuum $|0\rangle$ is the constant $\frac{1}{2}$ term, by removing it:

$$H|0\rangle = \sum_n \omega_n a_n^\dagger a_n |0\rangle = 0 |0\rangle \quad (7)$$

$$\begin{aligned} [H, a_n^\dagger] &= [\sum_{m=1}^{\infty} \omega_m a_m^\dagger a_m, a_n^\dagger] = \sum_{m=1}^{\infty} \omega_m [a_m^\dagger a_m, a_n^\dagger] \\ &= \sum_{m=1}^{\infty} \omega_m a_m^\dagger [a_m, a_n^\dagger] = \omega_n a_n^\dagger \end{aligned} \quad (8)$$

Therefore:

$$\begin{aligned} g_l &\equiv [H, (a_n^\dagger)^l] = [H, a_n^\dagger] (a_n^\dagger)^{l-1} + a_n^\dagger [H, (a_n^\dagger)^{l-1}] = \omega_n (a_n^\dagger)^l + a_n^\dagger [H, (a_n^\dagger)^{l-1}] \\ &= \omega_n (a_n^\dagger)^l + a_n^\dagger g_{l-1} = \omega_n (a_n^\dagger)^l + a_n^\dagger (\omega_n (a_n^\dagger)^{l-1} a_n^\dagger g_{l-2}) \\ &= 2\omega_n (a_n^\dagger)^l + (a_n^\dagger)^2 g_{l-2} = \dots \\ &= l\omega_n (a_n^\dagger)^l + (a_n^\dagger)^l g_0 = l\omega_n (a_n^\dagger)^l \end{aligned} \quad (9)$$

$$\begin{aligned} H|l_1, l_2, \dots, l_N\rangle &= H(a_1^\dagger)^{l_1} (a_2^\dagger)^{l_2} \dots (a_N^\dagger)^{l_N} |0\rangle \\ &= ((a_1^\dagger)^{l_1} H + l_1 \omega_1 (a_1^\dagger)^{l_1}) (a_2^\dagger)^{l_2} \dots (a_N^\dagger)^{l_N} |0\rangle \\ &= (a_1^\dagger)^{l_1} ((l_2 \omega_2) (a_2^\dagger)^{l_2} + (a_2^\dagger)^{l_2} H + l_1 \omega_1 (a_2^\dagger)^{l_2}) \dots (a_N^\dagger)^{l_N} |0\rangle \\ &= (a_1^\dagger)^{l_1} (a_2^\dagger)^{l_2} ((l_1 \omega_1 + l_2 \omega_2) + H) \dots (a_N^\dagger)^{l_N} |0\rangle = \dots \\ &= (a_1^\dagger)^{l_1} (a_2^\dagger)^{l_2} \dots (a_N^\dagger)^{l_N} (\sum l_i \omega_i + H) |0\rangle \\ &= \sum l_i \omega_i |l_1, l_2, \dots, l_N\rangle \end{aligned} \quad (10)$$

$$\begin{aligned} [\phi(\vec{x}), \phi(\vec{y})] &= \int \int \frac{d^3 p d^3 q}{(2\pi)^6} \frac{1}{2\sqrt{E_{\vec{p}} E_{\vec{q}}}} [a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}}, a_{\vec{q}} e^{i\vec{q}\cdot\vec{y}} + a_{\vec{q}}^\dagger e^{-i\vec{q}\cdot\vec{y}}] \\ &= \int \int d\mathcal{P} \left(e^{i(\vec{p}\cdot\vec{x} - \vec{q}\cdot\vec{y})} [a_{\vec{p}}, a_{\vec{q}}^\dagger] + e^{i(\vec{q}\cdot\vec{y} - \vec{p}\cdot\vec{x})} [a_{\vec{p}}^\dagger, a_{\vec{q}}] \right) \\ &= \int \int \frac{d^3 p d^3 q}{(2\pi)^3} \frac{1}{2\sqrt{E_{\vec{p}} E_{\vec{q}}}} e^{i(\vec{p}\cdot\vec{x} - \vec{q}\cdot\vec{y})} \delta(\vec{p} - \vec{q}) - \int \int \frac{d^3 p d^3 q}{(2\pi)^3} \frac{1}{2\sqrt{E_{\vec{p}} E_{\vec{q}}}} e^{i(\vec{q}\cdot\vec{y} - \vec{p}\cdot\vec{x})} \delta(\vec{q} - \vec{p}) \\ &= 0. \end{aligned} \quad (11)$$